

EX.NO : 12

IMPLEMENTATION OF GO BACK N PROTOCOL

AIM :

To write a Java program to perform a GoBackN Protocol.

ALGORITHM :

- 1) Start the program.
- 2) Prompt for the frame size.
- 3) Initialize sent to 0.
- 4) Transmit frames up to the frame size, printing each transmission.
- 5) Prompt for lost Acknowledgement.
- 6) If lost Acknowledgement equals frame size, stop; else, update sent and repeat transmission.
- 7) Stop the program.

PROGRAM :

```
import java.io.*;

public class GoBackN
{
    public static void main(String args[]) throws IOException
    {
        BufferedReader br = new BufferedReader(new
        InputStreamReader(System.in));

        System.out.println("Enter the Frame Size : ");
        int window = Integer.parseInt(br.readLine());
```

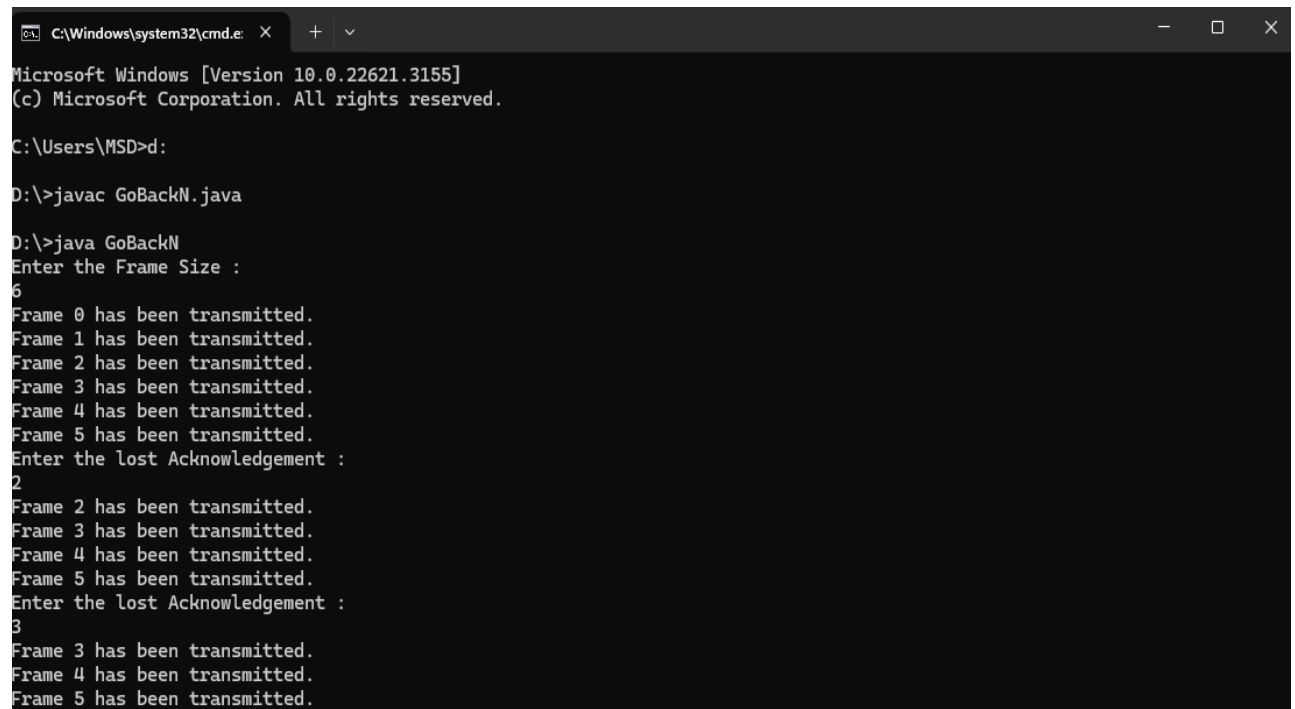
```
        boolean loop = true;
        int sent = 0;
        while(loop)
        {
            for(int i = 0; i < window; i++)
            {
                System.out.println("Frame " + sent + " has been
transmitted.");

                sent++;

                if(sent == window)
                    break;
            }

            System.out.println("Enter the lost Acknowledgement : ");
            int ack = Integer.parseInt(br.readLine());
            if(ack == window)
                loop = false;
            else
                sent = ack;
        }
    }
}
```

OUTPUT :



```
C:\Windows\system32\cmd.exe X + v
Microsoft Windows [Version 10.0.22621.3155]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MSD>d:

D:\>javac GoBackN.java

D:\>java GoBackN
Enter the Frame Size :
6
Frame 0 has been transmitted.
Frame 1 has been transmitted.
Frame 2 has been transmitted.
Frame 3 has been transmitted.
Frame 4 has been transmitted.
Frame 5 has been transmitted.
Enter the lost Acknowledgement :
2
Frame 2 has been transmitted.
Frame 3 has been transmitted.
Frame 4 has been transmitted.
Frame 5 has been transmitted.
Enter the lost Acknowledgement :
3
Frame 3 has been transmitted.
Frame 4 has been transmitted.
Frame 5 has been transmitted.
```

RESULT :

Thus, the program for Implementing a GoBackN Protocol is successfully executed and the output is verified.